

SET 2

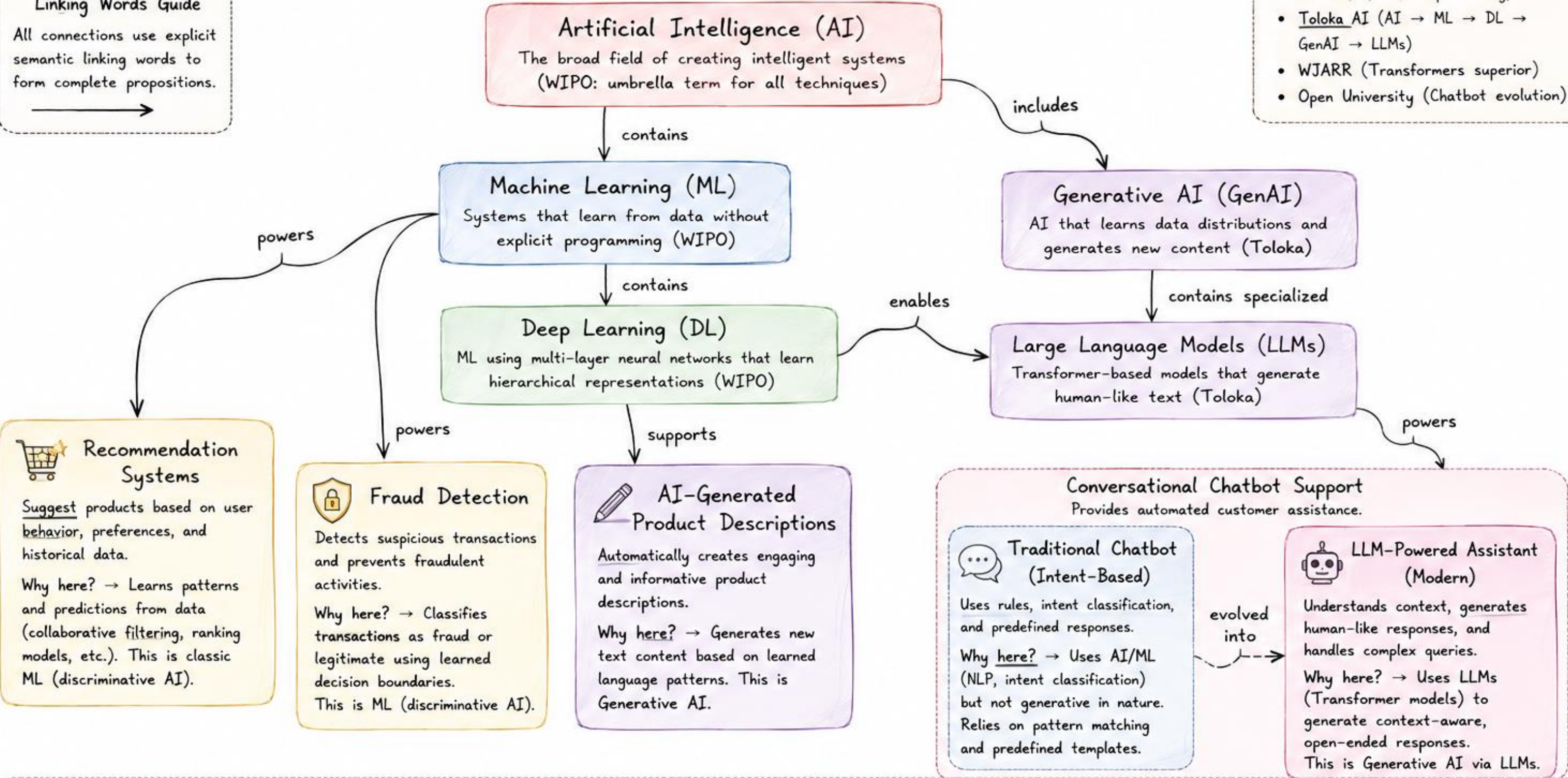
2. An e-commerce company uses recommendation systems, chatbot support, fraud detection, and AI-generated product descriptions. Develop a concept map illustrating which components belong to AI, ML, DL, Generative AI, and LLMs. Justify your classification.

5. Create a comparative mind map illustrating the differences among AI, ML, DL, Generative AI, and LLMs based on learning approach, data requirements, computational complexity, outputs, and applications. Recommend the best technology for educational chatbots.

E-Commerce Components Mapped to AI Taxonomy

Linking Words Guide
All connections use explicit semantic linking words to form complete propositions.
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- Taxonomy Source**
- WIPO (AI, ML, Deep Learning)
 - Toloka AI (AI → ML → DL → GenAI → LLMs)
 - WJARR (Transformers superior)
 - Open University (Chatbot evolution)



Justification Summary				
1. Artificial Intelligence (AI) The umbrella field that includes all techniques enabling machines to perceive, reason, learn, and act intelligently (WIPO).	2. Machine Learning (ML) Uses statistical methods to learn from data. Recommendation Systems and Fraud Detection require learning patterns and making predictions → ML (Toloka: Discriminative AI).	3. Deep Learning (DL) Subset of ML using neural networks with many layers to automatically learn complex features. It provides the foundation for modern generative models (WIPO).	4. Generative AI (GenAI) Focuses on generating new content (text, images, etc.) similar to training data. AI-Generated Product Descriptions belong here because they create new language outputs (Toloka).	5. Large Language Models (LLMs) Specialized GenAI models using Transformer architecture and self-attention. Power modern conversational assistants capable of rich, open-ended dialogue (Toloka, WJARR, Open University).

Literature Evidence: WJARR (2025) shows Transformers (basis of LLMs) achieve ~85% accuracy (vs ~65% for LSTM) in complex tasks. Open University study shows LLM chatbots provide better contextual understanding than traditional intent-based systems, though RAG is recommended for accuracy and safety.

Linking Words Guide

All connections use explicit semantic linking words to form complete propositions.

- Subset of AI that learns patterns from data without explicit programming (WIPO; Toloka AI).
- Focuses on making predictions or decisions (discriminative AI).



1. Artificial Intelligence (AI)



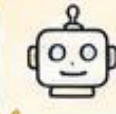
2. Machine Learning (ML)



3. Deep Learning (DL)



4. Generative AI (GenAI)



5. Large Language Models (LLMs)

- Specialized subset of Generative AI.
- Transformer-based models trained on massive text corpora to generate human-like language.
- Examples: GPT-4o, Claude, Llama 3, Gemini.

Comparative Mind Map AI, ML, DL, Generative AI & LLMs Key Differences

Taxonomy Sources

- WIPO (AI, ML, DL definitions)
- Toloka AI (AI → ML → DL → GenAI → LLMs)
- WJARR (Transformers superior to LSTM)
- Open University (Chatbot evolution)

- Subset/branch of AI that learns data distributions and generates new content similar to training data (Toloka AI).
- Produces text, images, audio, code, designs, etc.

Comparative Matrix

Dimension	AI	ML	DL	Generative AI	LLMs
Learning Approach	Broad intelligent systems using rules, logic, search, planning, or learning methods.	Learns patterns from data using statistical & optimization techniques.	Multi-layer neural networks learn hierarchical representations automatically.	Learns data distributions and generates new content similar to training data.	Transformer architecture with self-attention learns next-token prediction for language generation.
Data Requirements	Low to moderate; can work with rules or small datasets.	Moderate to large structured datasets (labeled).	Large datasets; often requires thousands to millions of examples.	Very large datasets (text, images, audio, video, code).	Extremely large text datasets (billions to trillions of tokens).
Computational Complexity	Low to very high depending on approach.	Moderate (CPU/GPU) depending on model.	High (GPUs/TPUs, parallel training, deep architectures).	High (large models, diffusion/GANs/Transformers require significant compute).	Very High (massive training compute, distributed/custees, high inference memory).
Primary Outputs	Decisions, rules, plans, control actions.	Predictions, classifications, rankings, recommendations.	Predictions, embeddings, feature representations.	New content: text, images, audio, video, code, designs.	Human-like text, conversations, summaries, explanations, translations, code, reasoning.
Representative Applications	Expert systems, robotics planning, rule-based automation.	Recommendation systems, fraud detection, spam filtering, forecasting.	Image recognition, speech recognition, autonomous driving, medical imaging.	AI-generated art, product descriptions, ad copy, synthetic data, design.	Chatbots, virtual assistants, tutoring, content creation, code generation, knowledge Q&A.

Recommendation for Educational Chatbots



Best Choice:

LLM-based system with Retrieval-Augmented Generation (RAG) and ML safety guardrails.

Why LLMs + RAG?

- ✓ Superior contextual understanding (WJARR: Transformers ~85% accuracy vs ~65% for LSTM in complex tasks).
- ✓ Handles open-ended Socratic dialogue and multi-turn conversations.
- ✓ RAG grounds responses in verified textbooks/notes, reducing hallucinations (Open University study).
- ✓ Scalable, adaptable, and supports personalized learning.

Recommended Hybrid Architecture

